

Machine Learning and Data Fusion Approach for Elastic Rock Properties Estimation and Fracturability Evaluation

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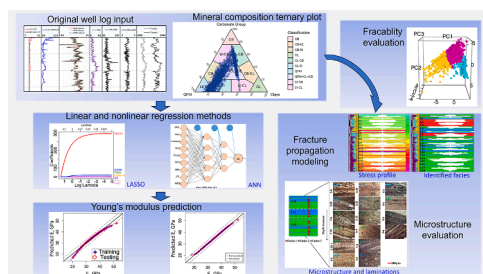
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HIGHLIGHTS

- Interdisciplinary approach merges data mining, experiments, and calibrated synthetics for improved elastic modulus predictions.
- Microstructural insights from rock images utilized for geomechanical facies analysis.
- Petrophysical data and well logs correlated with shear wave velocity (V_s) and Young's modulus.
- Machine-learning workflow predicts Young's modulus, assesses fracturability, incorporating mineral composition, geomechanics, and microstructure.
- Microscale thin section analysis and fracture modeling enhance fracture geometry understanding, aiding fracturability assessment.
- Demonstrates data mining, machine learning potential in predicting rock properties, assessing fracturability.

GRAPHICAL ABSTRACT



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ABSTRACT

Accurate rock elastic property determination is vital for effective hydraulic fracturing, particularly Young's modulus due to its link to rock brittleness. This study integrates interdisciplinary data for better predictions of elastic modulus, combining data mining, experiments, and calibrated synthetics. We used the microstructural insights extracted from rock images for geomechanical facies analysis. Additionally, the petrophysical data and well logs were correlated with shear wave velocity (V_s) and Young's modulus. We developed a machine-learning workflow to predict Young's modulus and assess rock fracturability, considering mineral composition, geomechanics, and microstructure. Our findings indicate that artificial neural networks effectively predict Young's modulus, while K-Means clustering and hierarchical support vector machines excel in identifying rock and geomechanical facies. Utilizing Microscale thin section analysis in conjunction with fracture modeling enhances our understanding of fracture geometries and facilitates fracturability assessment. Notably, fracturability is controlled by specific geomechanical facies during initiation and propagation and influenced by continuity of

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geomechanical facies in small depth intervals. In conclusion, this study demonstrates data mining and machine learning potential for predicting rock properties and assessing fracturability, aiding hydraulic fracturing design optimization through diverse data and advanced methods.

1. Introduction

In recent years, machine learning methods have been widely utilized in the petroleum industry for various applications. The integration of multi-stage hydraulic fracturing and horizontal drilling has unlocked the potential of unconventional reservoirs by creating intricate networks of highly conductive fractures in the rock matrix [[2], [22], [23]]. Accurate knowledge of the rock's mechanical properties is essential for designing efficient fracturing operations. However, estimating precise elastic moduli for thin-bedding formations like shale remains challenging due to the heterogeneity that complicates obtaining high-quality continuous cores [[16], [18], [24]]. Moreover, obtaining a substantial number of measurements to avoid biased results can be cost-prohibitive [4]. Consequently, the accurate estimation of elasticity moduli has garnered continuous attention from geologists, petroleum engineers, geophysicists, and petrologists over the past decade.

Despite deep insights into geological characteristics, rock mechanical properties, and formation stress states, the propagation of fractures and their interaction with pre-existing microcracks remain enigmatic and intricate to evaluate. Laboratory experiments on rock deformation have indicated that macroscopic fracture generation follows the initiation, development, and merging evolution of microcracks [11]. With altered in-situ stresses after fracturing, microcracks locally become activated and subsequently coalesce into macroscale fractures, leading to an abrupt buildup of absolute cumulative energy [15]. In essence, both microscopic features and macroscopic factors contribute to hydraulic fracture initiation, propagation, and the development of fracture systems. Zhang et al. [26] and Li et al. [8] believe that the brittleness of the rock is highly correlated to the failure behavior of the rock so that the fracturing behaviors of the rock can be equivalent to the brittleness characteristics. These studies are commonly referred to as fracturability evaluation. However, previous studies typically limited themselves to assessing rock brittleness, considering fracturability as equivalent to rock brittleness. Nonetheless, rock fracturability is intricately linked to parameters governing fracture propagation, such as stress, rock elastic properties, anisotropic behavior, microstructure development, and

more. Thus, there remains a significant gap in effectively integrating diverse data across varying scales and perspectives into a cohesive framework for fracturability evaluation, necessitating robust methods or workflows to address such intricate challenges.

The domain of machine learning has found substantial application in the petroleum industry in recent times. The two primary learning methods are supervised learning, encompassing regression and classification, and unsupervised learning, which employs clustering techniques. In supervised learning, these methods have been employed for assessing unknown reservoir properties. Notably, Wei et al. [25] predicted rate-dependent compressive strength of rocks, Gong et al. [4], [5] successfully forecasted Young's modulus, and Gupta et al. [6] and Niu et al. (2021) utilized machine learning for production forecasting. Unsupervised learning applications encompass rock typing and brittleness evaluation, as demonstrated by Tran N. et al. [20] and microseismic-based rock typing by Alzate and Deveglozda [1]. Lubbers et al. [13] created a Machine Learning surrogate model to capture adsorption effects in porous media across a broad parameter range, bridging the molecular dynamics (MD) and Lattice Boltzmann Methods scales with minimal MD calculations. Additionally, a third common machine learning approach revolves around enhancing performance for parameter tuning objectives, exemplified by Liu et al. [9] for wellhead treating pressure matching and Liu et al. [10] for production history matching. Leveraging the adaptability of machine learning, the petroleum industry continues to elevate its operational efficiency through more accurate predictions, insightful characterizations, and improved parameter optimization.

This study aims to apply data mining principles to establish a deeper understanding of the relationship between rock characteristics, elastic properties, microstructure, and rock fracturability. More specifically, the objective of this work is to evaluate Young's modulus of the rock using a broader dataset/feature, and to evaluate the fracturability of the rock using our proposed new approach. The remainder of this work is organized as follows: the methodology section outlines the wide range of data sources, including well log data, mineral composition data and their roles to the mechanical rock properties estimation. This section is followed by

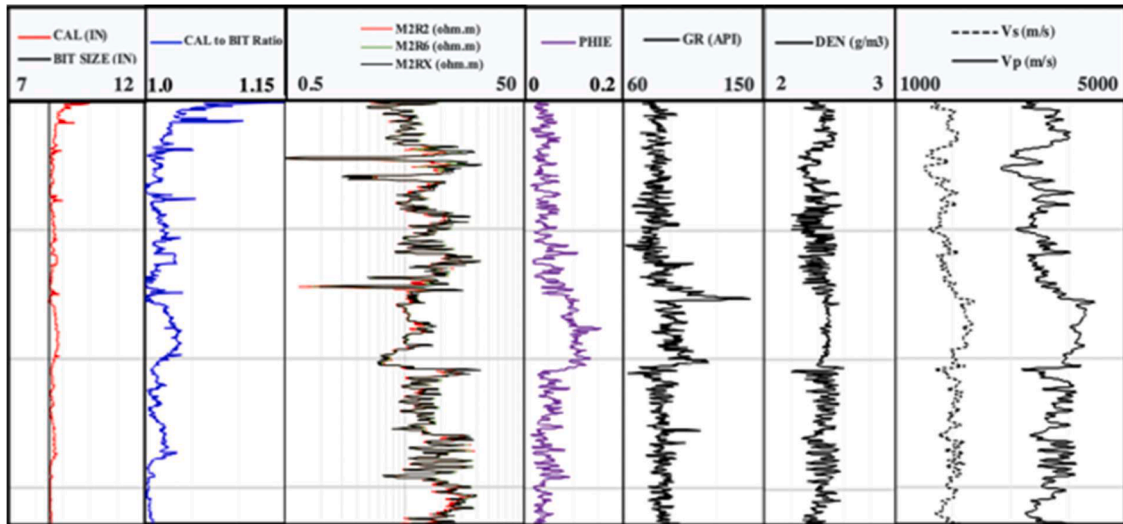


Fig. 1. Original well log input. The well log in each track are as follows: first track: caliper log measurement (CAL) and drill bit size (BIT SIZE); second track: the ratio between caliper log measurement to drill bit size; third track: resistivity log of shallow (M2R2), intermediate (M2R6) and deep (M2RX) measurement; fourth track: effective porosity (PHIE); track five: gamma ray log (GR); track six: density log (DEN); track seven: compressive velocity (v_p) and shear velocity log (v_s).

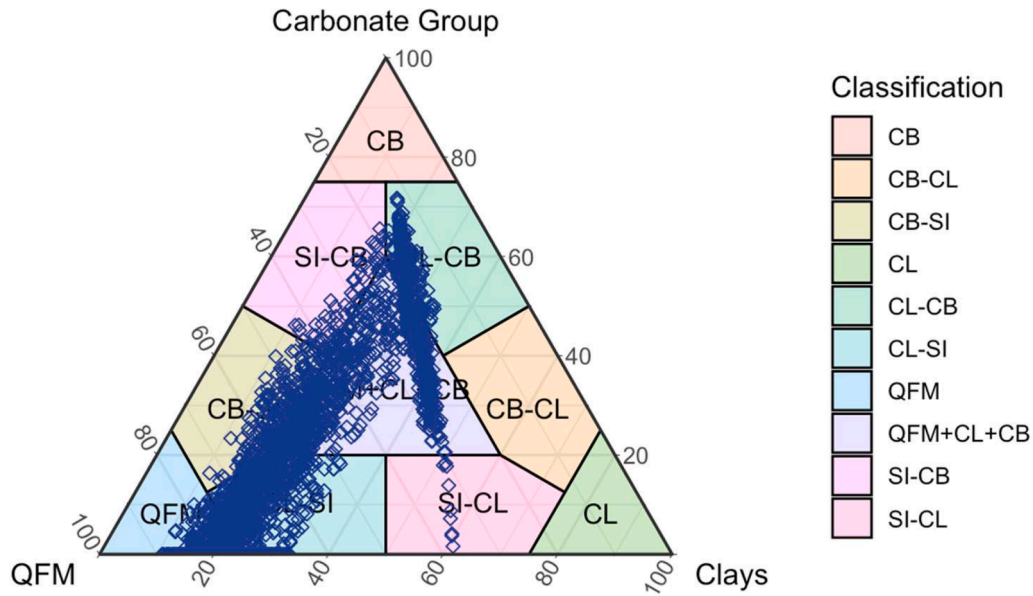


Fig. 2. Mineral composition ternary plot of the input dataset. The mineralogy composition ternary plot is split to ten regions based on the mineral composition group characteristics. In here, CB, CL, SI refers to carbonate group, clay group, silicate mineral group (quartz and feldspar and mica). Most classifications are the combination of two general groups as indicated in their names connected by the hyphen in the legend. For example, CB-CL stands for carboniferous clay rock while CL-CB refers to clayey carbonate rock.

the introductions of the machine learning techniques that are involved in feature selection process, regression process (Young's modulus estimation), and clustering process (fracability analysis). The results section presents regression outcomes of Young's modulus and the comparisons of different machine learning approaches are also presented. Next, the fracability analysis results are presented compared from different machine learning approaches. Finally, the conclusion section summarizes the main findings and recommendations of this work.

2. Dataset, Methodology and Workflow

Section 2 consists of three major parts, including the dataset descriptions and preparation introduction in section 2.1, major machine learning algorithms involved in this study in section 2.2, and the proposed workflow explanation in section 2.3.

2.1. Data Sources and Preparation

The studied well is located in the same sag of Bohai Bay Basin in the northeast of China. The original well log data includes caliper log (CAL), shallow resistivity (M2R2), intermediate resistivity (M2R6) and deep resistivity (M2RX) log, bulk density (DEN or RHOB), effective porosity log (PHIE), gamma-ray log (GR), the compressional velocity log (v_p), and shear velocity log (v_s) (Fig. 1). The studied interval of the well is 344.5 m, including 2757 data points.

The availability of static mechanical rock properties from lab measurements, such as Young's modulus and Poisson's ratio, is usually limited. In our study, the geomechanical properties are thoroughly studied based on the original well log data and well-calibrated using lab measurement through the geomechanics study.

The mineral composition is highly relevant to the pore structure features and permeability variations in the shale rock (Xiong et al., 2019); hence it is employed as a group of the most important dataset. The mineral composition model was received as a continuous profile, which was interpreted from corresponding mineralogy logging and was calibrated by X-Ray Diffraction (XRD) lithology from laboratory studies. Fig. 2 illustrates the mineral composition distribution on a ternary plot of the target layers. The mineral composition ternary plot reflects the depositional histories and covers over 95% of the mineral compositions;

it also honors the mechanical and chemical response affinities of each group ([12],[28]).

Regarding the microstructure, we employed thin rock sample slides capable of capturing microstructures at a scale of a few hundred microns. A set of essential reservoir conditions—comprising stress state, the presence of natural fractures, operational parameters, and mechanical rock properties—constitutes the factors that define fracability or the efficiency of hydraulic fracturing.

We account for the mechanical anisotropic nature by comparing Young's modulus measured at a 0-degree phase angle to that at a 90-degree phase angle (denoted as “E0toE90” in the rest of the paper). Furthermore, we utilize rock thin-section images as an additional tool to aid our interpretation of anisotropy.

Conventionally, the dipole sonic shear velocity splits into fast shear velocity and slow shear velocity, indicating the degree of fracture intensity. A wider split signifies greater variation in fracture intensity, ranging from microfractures to higher-degree fractures. However, in vertical wells, the difference between fast and slow shear velocities tends to be negligible. Consequently, since our focal well is a vertical exploration well, we cannot integrate the shear velocity splitting technique into this study. Additionally, core samples scanned using computed tomography (CT) typically provide insights into whether the fast and slow shear velocities stem from bedding planes or result from lamination and fractures. As such, we intend to leverage core sample images, along with thin-section images, to provide supplementary support for our study.

2.2. Regression and Clustering Involved Algorithms

LASSO, Elastic Net and Ridge are three regularization algorithms, which can be used to select essential features among the overall input features by coefficient shrinking process. We developed the cross-disciplinary data usage workflow which requires assessment of the importance of data attributes, thus, we will use these algorithms to evaluate the input features importance.

LASSO stands for Least Absolute Shrinkage and Selection Operator, first established by R. Tibshirani [19]. One of its main objectives is deploying the shrinking process or the regularization process, to penalize a portion of the feature coefficients approaching zero. The

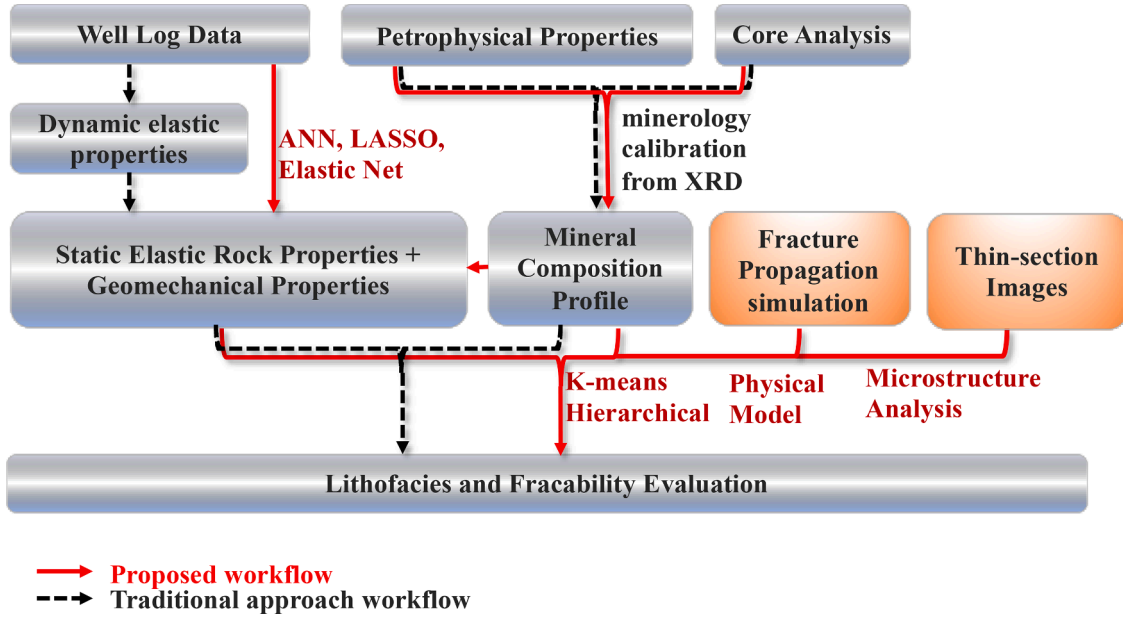


Fig. 3. Traditional workflow and the proposed workflow.

parameter lambda in equation (1) controls the penalty extent. The greater value of lambda increases the penalty strength to force some of the feature coefficients to zero. This process reduces the model variance without introducing large bias, leading to reduced feature numbers or decreasing dimensions. When some features in the input dataset are highly correlated, LASSO elects one among all and discards other highly correlated features.

Usually, the two critical components of statistics, bias and variance, should be considered. The model prediction error consists of errors from large variance, errors from great bias, and the unexplained remainder. The OLS (Ordinary Least Square) intrinsically shows good unbiased characteristics. In LASSO, the penalty term is added to the non-zero coefficients as a form of L1 penalty showing in the second term in equation (1)

$$L_{LASSO}(\hat{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i' \hat{\beta})^2 + \lambda \sum_{j=1}^m |\hat{\beta}_j| \quad (1)$$

where y_i is the actual output of a single realization, $\hat{\beta}$ is the coefficient array, and $\mathbf{x}_i$ is the matrix of the input variables.

The Ridge as an L2 regularization method has a second order penalty term for the non-zero coefficients.

$$L_{Ridge}(\hat{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i' \hat{\beta})^2 + \lambda \sum_{j=1}^m \hat{\beta}_j^2 \quad (2)$$

As an extension, Zou and Hastie [29] proposed Elastic Net as an extension of form of L1 and L2 regularizations. The objective of elastic net uses a mixed fashion of penalty terms from both ridge and the LASSO by minimizing the following lost function:

$$L_{ELnet}(\hat{\beta}) = \sum_{i=1}^n \frac{(y_i - \mathbf{x}_i' \hat{\beta})^2}{2n} + \lambda \left(\frac{1-\alpha}{2} \sum_{j=1}^m \hat{\beta}_j^2 + \alpha \sum_{j=1}^m |\hat{\beta}_j| \right) \quad (3)$$

where α is the weighting parameter, usually set as 0.5.

Subsequent to data collection and preparation, we subjected the data to dimension reduction techniques like LASSO, Ridge, and Elastic Net for cross-comparisons. In the context of Young's modulus estimation, we examined five distinct algorithms: three linear models, including LASSO, Ridge, and Elastic Net, and two non-linear models, including ANN, and MARS. We adopted this combination of linear and non-linear

prediction approaches since LASSO, Ridge, and Elastic Net also function as feature selection algorithms, diminishing the dimensionality of input attributes. Given the uncertainty regarding the linearity of the multi-variate regression within the dataset, we gauged each method's performance through error analysis.

Conversely, the clustering and fracability assessment phase integrated K-Means clustering, with Hierarchical clustering bolstered by Support Vector Machine. The utilization of Hierarchical clustering in tandem with Support Vector Machine was prompted by the need to construct a classifier for prediction, given Hierarchical clustering's inability to partition the high dimensional data. The dataset was partitioned, allocating 80% for training and 20% for testing. Owing to the dataset's intricate nature involving a considerable number of feature attributes, practical considerations necessitated data scaling or normalization.

2.3. Proposed Workflow

Fig. 3 outlines the conventional workflow of the typical integrative petrophysics and geomechanics research in the dashed-line paths together with the proposed workflow of this study in red-line paths. For the proposed workflow, we provide an additional aspect using cross-disciplinary data. We summarized the superior advantages and characteristics of the proposed workflow as compared to the traditional approach in the following four aspects. First of all, the significance of the proposed workflow is avoiding static-to-dynamic conversion. The different frequency and strain amplitudes are usually considered the reason causing the difference between the static and dynamic elastic property measurements [14]. Empirical correlations are employed converting the dynamic profile to the static in the absence of the static measurements. However, these correlations are generally rock-type dependent and are sometimes field-dependent too [17]. Thus, our workflow develops a robust approach to avoid specifying the rock type nor relying on consolidated dynamic-static conversion. The static elastic property estimations from our approach account for the mineral compositions and rock type while the essential features are selected on the basis of both physical reasoning and insights from the data itself. Secondly, we have incorporated machine learning method to find the intrinsic relationship between the acoustic well log data, mineral composition to the static rock properties by using the ANN, LASSO, and Elastic Net regression methods. Thirdly, the lithofacies and fracability

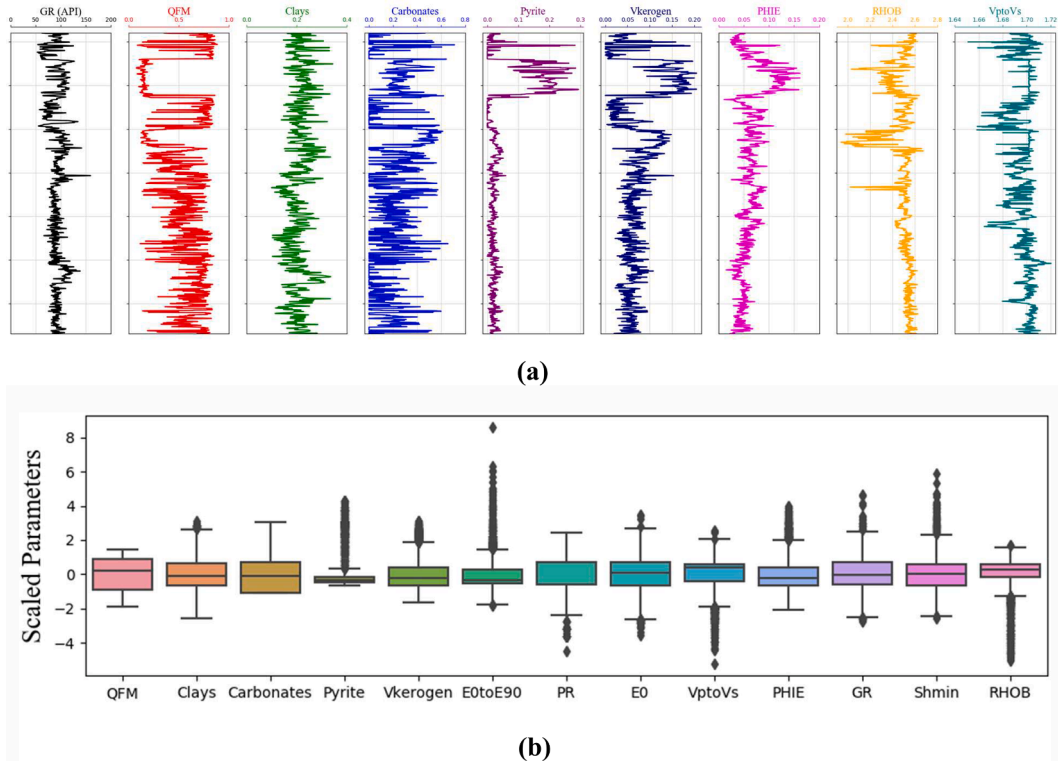


Fig. 4. Original attribute in the box plot of scaled candidate attributes for Young's modulus estimation in (a) well log form and in (b) box plot form. From left to right, the original candidate attributes of the model includes: quartz / feldspar / mica group (QFM), clay group (Clays), carbonate group (Carbonates), Pyrite, total organic content (TOC or Vkerogen), Young's modulus measured at 0 degree phase to its measured at 90 degree (E0toE90) as anisotropic indicator, Poisson's ratio (PR), Young's modulus measured at 0 degree (E0), ratio of compressive velocity to shear velocity (VptoVs), effective porosity (PHIE), gamma ray log (GR), minimum horizontal stress (Shmin), density log (RHOB).

evaluation is conducted by machine learning clustering approach automatically. This practice allows us to evaluate the fracability from a boarder range of the data inputs systematically. Lastly, we incorporated the additional data and validation process to rigorously evaluate the lithofacies and fracability of the rock, including the fracture propagation numerical simulation and thin-section image microstructure analysis. This practice ensures the confidence of the lithofacies and fracability evaluation results.

The cross-disciplinary well logs in target interval and their box plots of all attributes before the dimension reduction is presented in Fig. 4 (a) and Fig. 4 (b) respectively. From this figure, we noticed that some attributes, such as E0toE90, VptoVs, and Pyrite, have notable numbers of outliers. Most attributes show desirable variabilities. In order to identify the most significant features for estimating Young's modulus (denoted as E0), the dimension reduction technique should be applied. This process is discussed in the E0 estimation section.

3. Results

Section 3 is organized into two major sub-sections, including the result of Young's modulus estimation using regression methods in section 3.1 and the fracability analysis using both clustering methods, numerical simulations, and microscopic images in section 3.2.

3.1. Young's Modulus Estimation

The original dataset includes the following attributes: QFM, Clays, Carbonates, Pyrite, Vkerogen, E0toE90, PR, E0, VptoVs, PHIE, GR. Fig. 5 includes the coefficient plot versus tuning parameter lambda of both LASSO and the Elastic Net, and Ridge. From the table, the higher the coefficients, the more significant impact it serves in the regression methods. From the upper row of LASSO, we can notice that the top

essential features determined by greater coefficients in the plot are VptoVs (ratio of compressional velocity to shear velocity), PHIE (effective porosity), RHOB (bulk density), clays and pyrite, PHIE (effective porosity), Pyrite, and Clays (clay content).

In Fig. 5b, the cross-plot of MSE to logarithmic values of tuning parameter lambda suggested two options of "optimal" solutions. They are the lambda that gives the minimum mean cross-validated error (λ_{minMSE}) and the most regularized model such that error is within one standard error of the minimum (λ_{1se}). From the mean squared error plot, the two λ 's marked out with dash line are λ_{minMSE} and λ_{1se} , respectively. From the following three models, we noticed that their coefficient plots are quite similar in both LASSO and the elastic net, while the Ridge shows a slight difference from the other two. Table 11 summarizes the λ_{minMSE} and λ_{1se} , in addition to the number of suggested features from each λ 's. In other words, the top ranked features are selected from each algorithm at its λ_{1se} . (Table 1)

All regression coefficients of six regression criteria from three methods are summarized in Table 2. When looking at three minimum MSE regressions from three methods, we rank the features as follows: VptoVs, bulk density, porosity, Clays/Pyrites, and GR. In regressions from λ_{1se} , the top rankings are VptoVs, bulk density, porosity, Clays/or GR. This explains the inversion process of estimating the Young's modulus from the physical model as most physical model uses acoustic logs and bulk density, and lithology indicators. In addition, clays and pyrites are two important mineral groups. Usually, the clay group harms the stiffness of the rock therefore is negatively related to Young's modulus.

The regression performance is presented by the cross-plot of predicted Young's modulus to the actual values of Young's modulus, the residual plots of each regularized, and their root mean square error (RMSE) in Fig. 6. We observed promising agreements in the middle

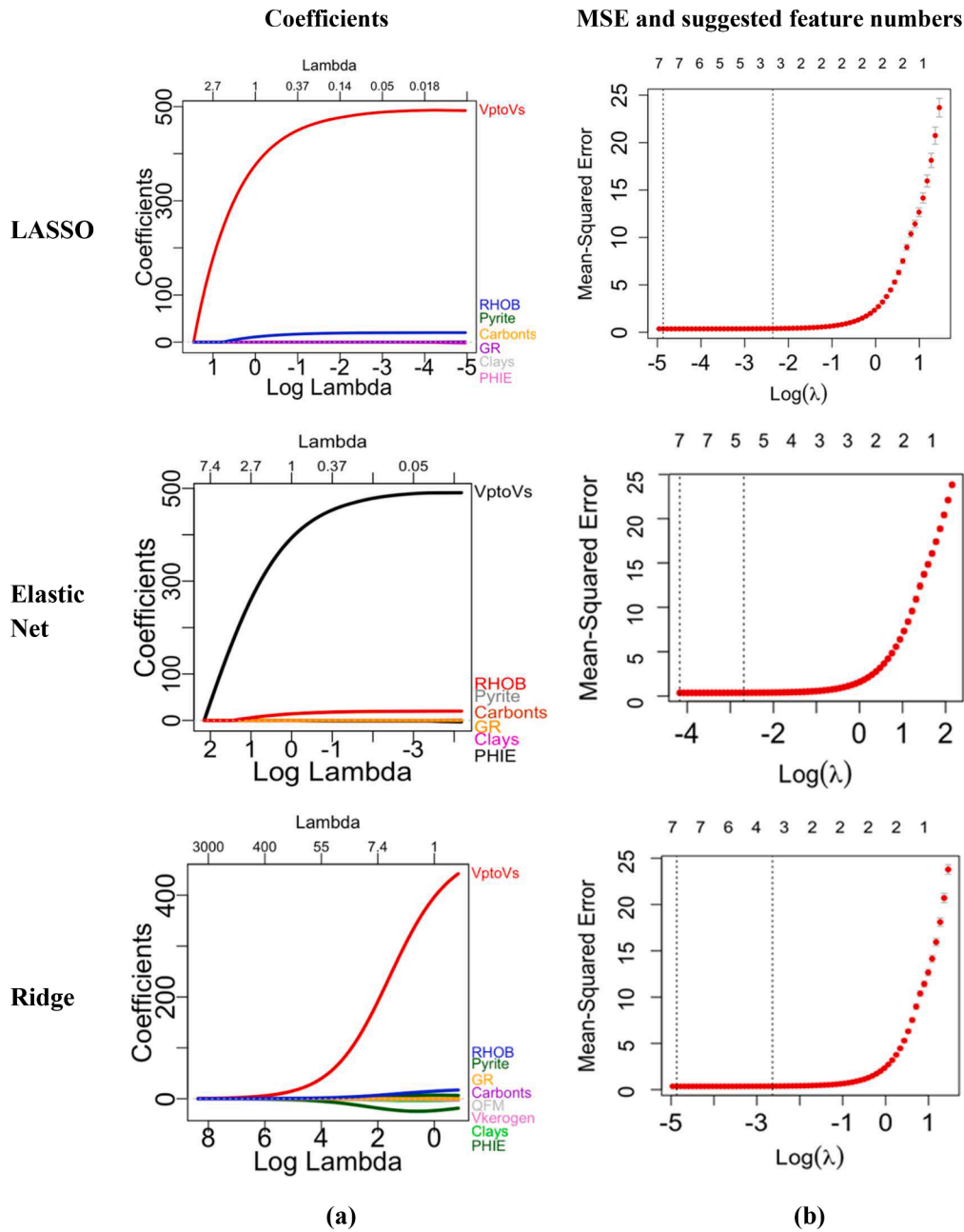


Fig. 5. (a) lambda plot and the (b) mean square error plot of LASSO, elastic net and Ridge regularization methods.

Table 1

The λ values giving the two best solution scenarios.

	$\lambda_{\min\text{MSE}}$	Suggested feature numbers from $\lambda_{\min\text{MSE}}$	$\lambda_{1\text{se}}$	Suggested feature numbers from $\lambda_{1\text{se}}$
LASSO	0.0077	7	0.095	3
Elastic Net	0.015	7	0.068	5
Ridge	0.0070	7	0.086	4

range of the output. Three algorithms all underestimated Young's modulus at both higher-end and lower-end ranges. This should be a sign that the input features to the output Young's modulus disobeys a simple first order linear relationship.

To explore the non-parametric non-linear algorithm, we employed both ANN and MARS as comparisons to linear regularized approaches. In ANN, to honor the full capacity of ANN, we tested both ANN with all

Table 2

The sparse matrix of the selected attributes in LASSO, Elastic Net and Ridge.

Sparse matrix parameters	LASSO $\lambda_{\min\text{MSE}}$	Elastic Net $\lambda_{\min\text{MSE}}$	Ridge $\lambda_{1\text{se}}$	LASSO $\lambda_{1\text{se}}$	Elastic Net $\lambda_{1\text{se}}$	Ridge $\lambda_{\min\text{MSE}}$
(Intercept)	-848.13	-846.06	-848.13	-831.03	-838.54	-836.77
QFM	.	.	.	.	.	.
Clays	-0.82	-0.87	-0.82	.	-0.30	.
Carbonates	0.17	0.16	0.17	.	.	.
Pyrite	0.82	0.96	0.82	.	.	.
Vkerogen	.	.	.	.	.	.
VptoVs	491.59	490.51	491.59	481.90	486.38	485.12
PHIE	-2.64	-3.11	-2.64	.	-1.18	.
GR	-0.01	-0.01	-0.01	-0.002	-0.01	-0.002
RHOB	20.13	20.05	20.13	19.52	19.69	19.70

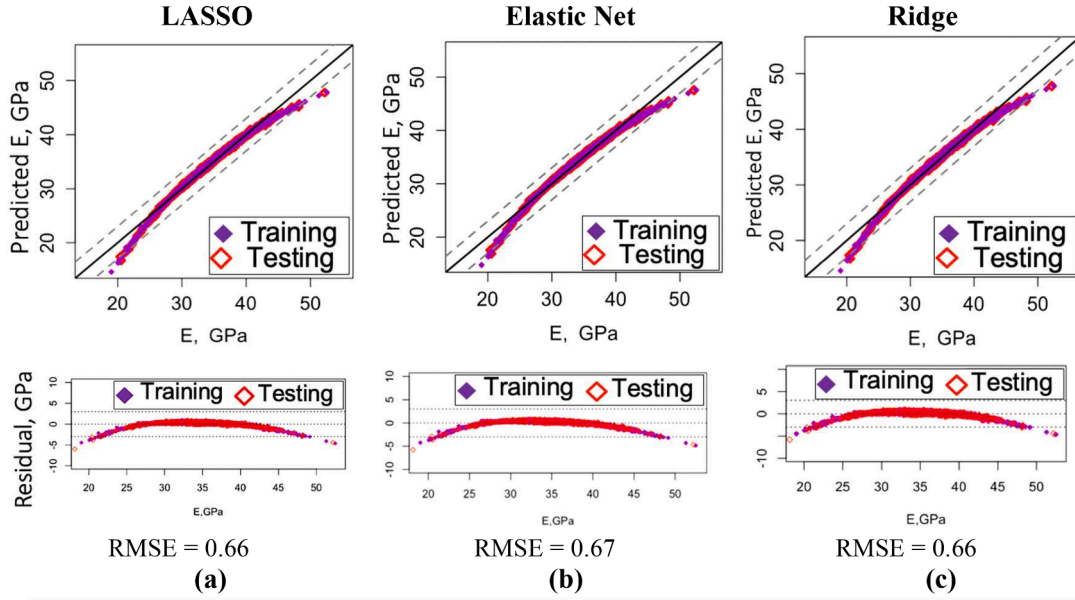


Fig. 6. Regression performance from linear based regularized regression models of (a) LASSO model (b) Elastic Net model, and (c) Ridge model.

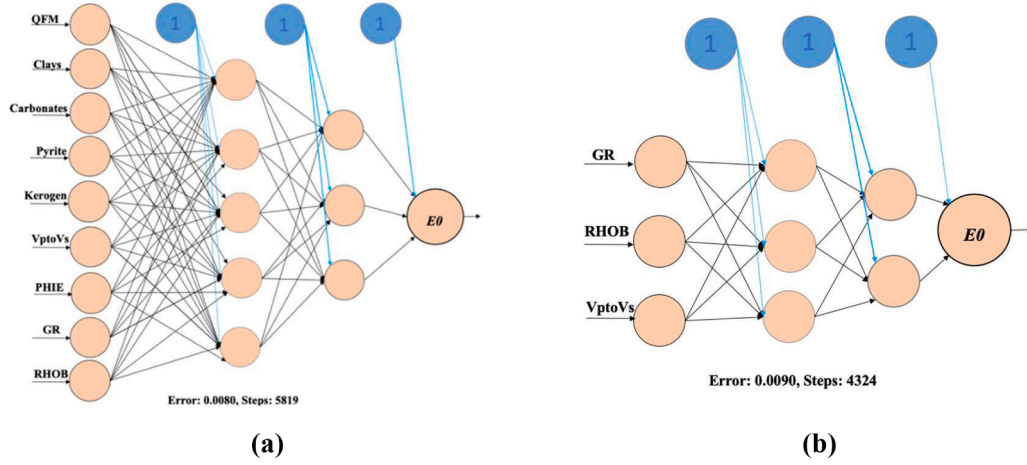


Fig. 7. ANN structure of (a) original all features and (b) the reduced features from LASSO feature selection.

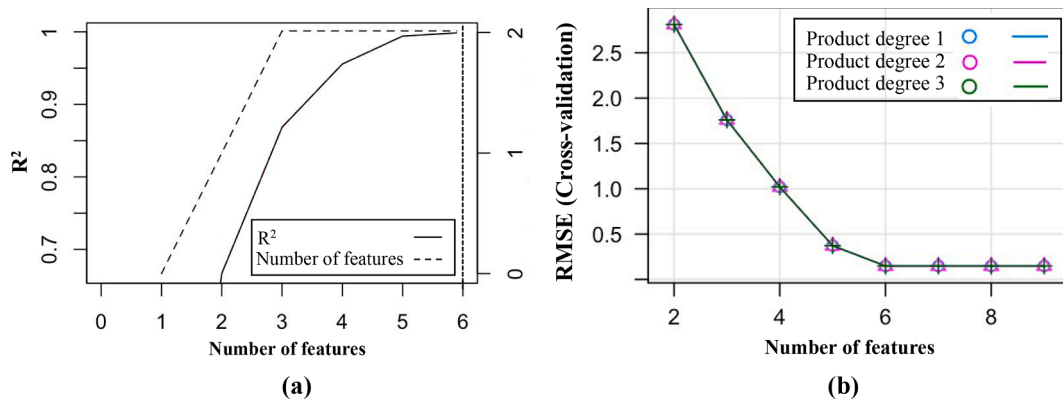


Fig. 8. MARS (a) single degree and (b) modeling tuning process.

original features (Fig. 7a) and reduced features from LASSO feature selection (Fig. 7b). Noticing in Fig. 7, we included dummy nodes indicated in blue in each of the hidden layers and output layer. To obtain

multiple classes with one neural network, we need to include the dummy variable (indicated in blue nodes in Fig. 7) for each class in each hidden layer. However, to optimize the prediction when the data is more

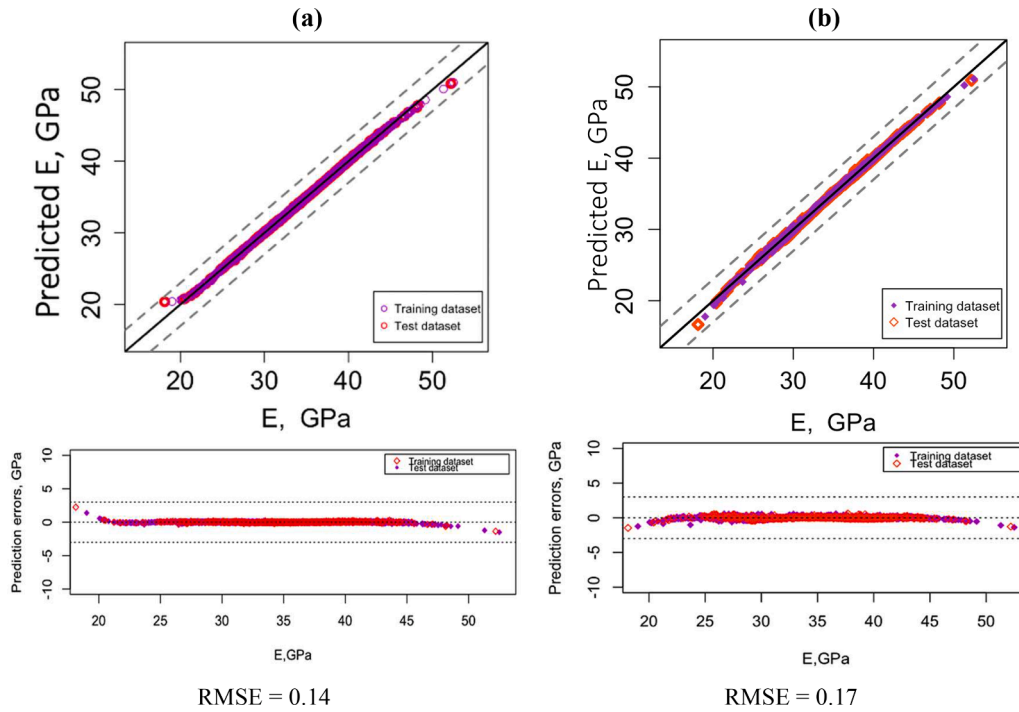


Fig. 9. Performance of non-linear regression models with 1:1 actual to prediction plot in the upper row and residual plot in the bottom row for (a) ANN and (b) MARS.

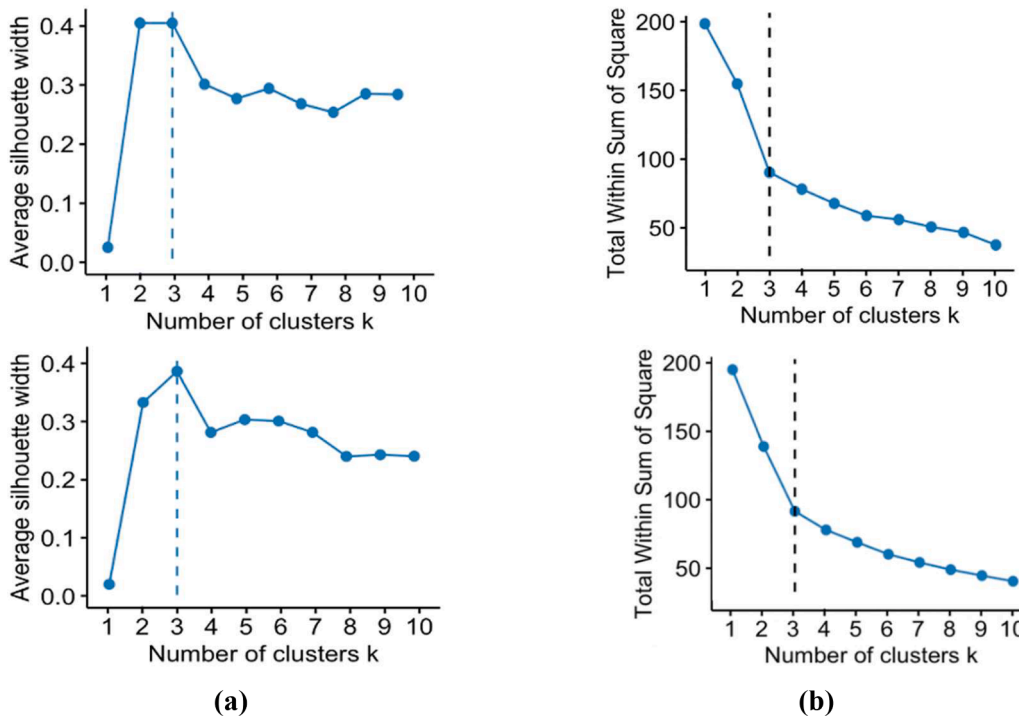


Fig. 10. (a) The silhouette width criterion and (b) the total within the sum of square criterion to determine the number of the cluster. The upper two plots are silhouette width against the number of clusters using *k*-means while the lower two are those if using hierarchical clustering.

Table 3

The number of predicted realizations of each facies of training dataset.

	Facies 1	Facies 2	Facies 3
<i>k</i> -means	417	1281	507
Hierarchical	532	1149	524

complex, other optimization algorithms, such as gradient descent, Genetic Algorithm, Tabu Search, is applied together with ANN to optimize the prediction by tuning the structures of the model. In this study, the accuracy is good enough so that there is no need to continue the ANN structure optimization.

For non-linear regressions, some of them, such as MARS (Multivariate Adaptive Regression Splines) have a self-feature selection scheme so

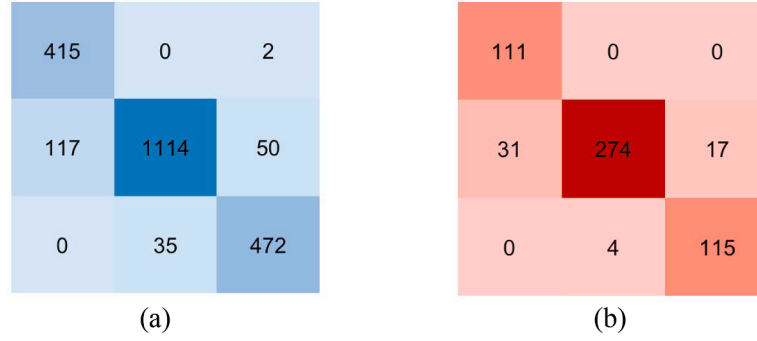


Fig. 11. Hybrid-cross-validate matrix of (a) training set and (b) test set. These plots are designed in the fashion of the confusion matrix of observation and prediction. In here, the rows refer to k-means, while the columns refer to hierarchical clustering.

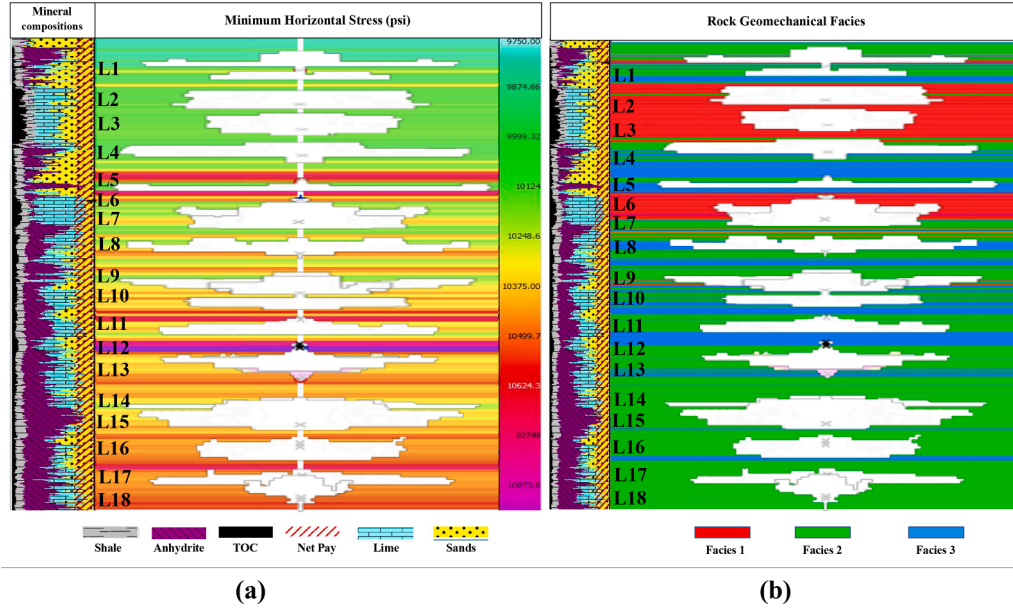


Fig. 12. The vertical profile of (a) minimum horizontal stress in the middle part (the stress is colored in the middle column; the color bar of the stress is indicated on the right; we consider the minimum horizontal stress is one of the dominant factors of fracture propagation) with rock mineral composition profile on the left and the (b) geomechanics facies (predicted by the clustering approach). The numerical simulation results of fracture geometries from fracture modeling software GOHFER are overlaid on both minimum horizontal stress profile and geomechanics facies profile.

that we do not need to reduce the features beforehand. MARS is a famous multivariate nonparametric regression method which evaluates the cut points or the knots with certain affinity to step functions. It provides more flexibility of the input features allowing both class-based parameters and numerical parameters. A noticeable difference from linear models is that it doesn't require normalization of the input parameters. To explore this model, we created a tuning matrix that accessed the degree of interaction complexity and the number of retained terms (Fig. 8). MARS shows that 6 of 6 terms of manipulated features from original ones and 2 of 9 predictors of original features are selected. The only two selected predictors are the VptoVs and the RHOB while all 7 others from the original attributes are not used. The model termination at the RMSE reaches 0.0001 at 6 terms. From Fig. 8(b), it indicates that three degrees of interaction perform the same on RMSE, and 6 terms of features reaches the best performance of RMSE.

The prediction performance on a 1:1 plot, residual plots of prediction for both ANN and MARS, and their RMSE are presented in Fig. 9. They both show improved prediction accuracies. From the linear regularization approaches (Fig. 6), we noticed that to avoid bias issues, the models yield lower estimations at lower bound and upper bound regions while the prediction performance at intermediate ranges is slightly

overpredicting. For the non-linear methods of ANN and MARS, the residuals plots both show improved bias balance and accuracy.

Based on the performances of the above 5 methods, the regularized approaches of the first three show promising performance while ANN and MARS have relatively smaller RMSE than linear regularized methods. Among all, ANN yields the best performance. To have an even more rigorous analysis, hyper parameter tuning of the learning rate of the regularization approaches, ANN hidden layer numbers, number of nodes in each hidden layer, number of degrees, and number of predictors are suggested. In our model, the regression performance is good enough from regularization and ANN without tuning.

3.2. Fracability Analysis Through Clustering and Numerical Simulation

Usually, the rock elastic properties, together with geomechanical properties are used for part of the comprehensive reservoir characterization studies and will be used for selecting landing point, thereafter, following the well completion design. An important factor controlling the vertical landing point and the vertical completion strategies is the potential stimulated reservoir volume from an ensemble of fractures in a horizontal well, which is largely controlled by so called fracability. Thus,

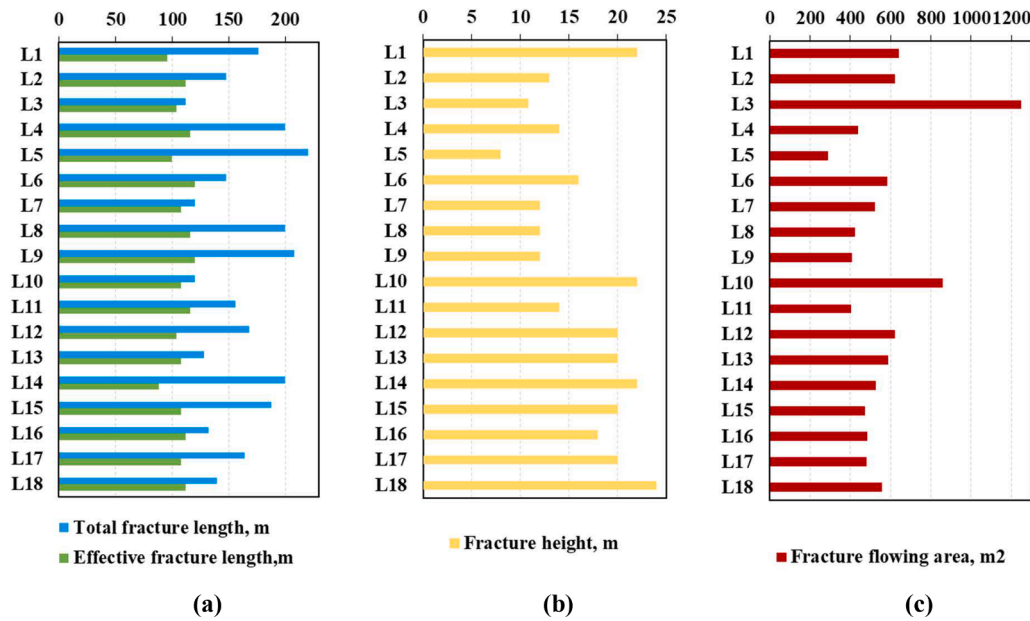


Fig. 13. The fracture geometries from GOHFER fracture modeling of the studies interval. (a) the total fracture half-length and the effective fracture half length. (b) Fracture height. (c). Fracture flowing area.

our next step is to investigate the geomechanical facies to guide the fracability through clustering of facies together with our analysis. The fracability facies are built based on the rock lithology well logs, mineral composition, and the results of comprehensive geomechanics study. We believe the mineral composition, lithology related well logs, and geomechanical properties have close ties to the geomechanical groups of the rock. The identified clusters from aforementioned input parameters will ultimately be defined as geomechanical facies.

Clustering is an unsupervised learning method that explores the nature of the data itself. However, the number of clusters must be determined beforehand. Fig. 10 presents two criteria for determining the number of clusters based on the silhouette width and the total within the sum of squares (WSS) of the number of clusters. From the average silhouette method, the 3-cluster leads to the maximum average silhouette width as the optimal suggested cluster number from the algorithm. Whereas the elbow method uses a rule of thumb where the users determine the “elbow” of the curve on the within sum of square (WSS) against number of clusters plot. The goal of this determination is that the optimal number of cluster k is selected without excessive and redundant interpretations representing the whole data set in terms of total within sum of square. From WSS against k plot, we found that the ‘elbow’ of the curve is close to 3. This avoids minimal incremental WSS explained by the cluster nor losing essential WSS of the whole dataset. From both k -means and hierarchical clustering of WSS and the silhouette method suggests 3 cluster as the best scenario.

In this work, the clustering algorithms are not compared but used as cross-validation to determine the fracability/ geomechanical facies. The objective of the study is to investigate the fracable characteristics of the rocks from the geomechanics perspective and the pre-existing fractures perspectives (both micro and macro scales) as they are the main physical controls of the success of the fracturing job. The geomechanical properties, such as anisotropy, brittleness, and the pore shapes, are considered relevant parameters hence to be included in the study for the preliminary inspections.

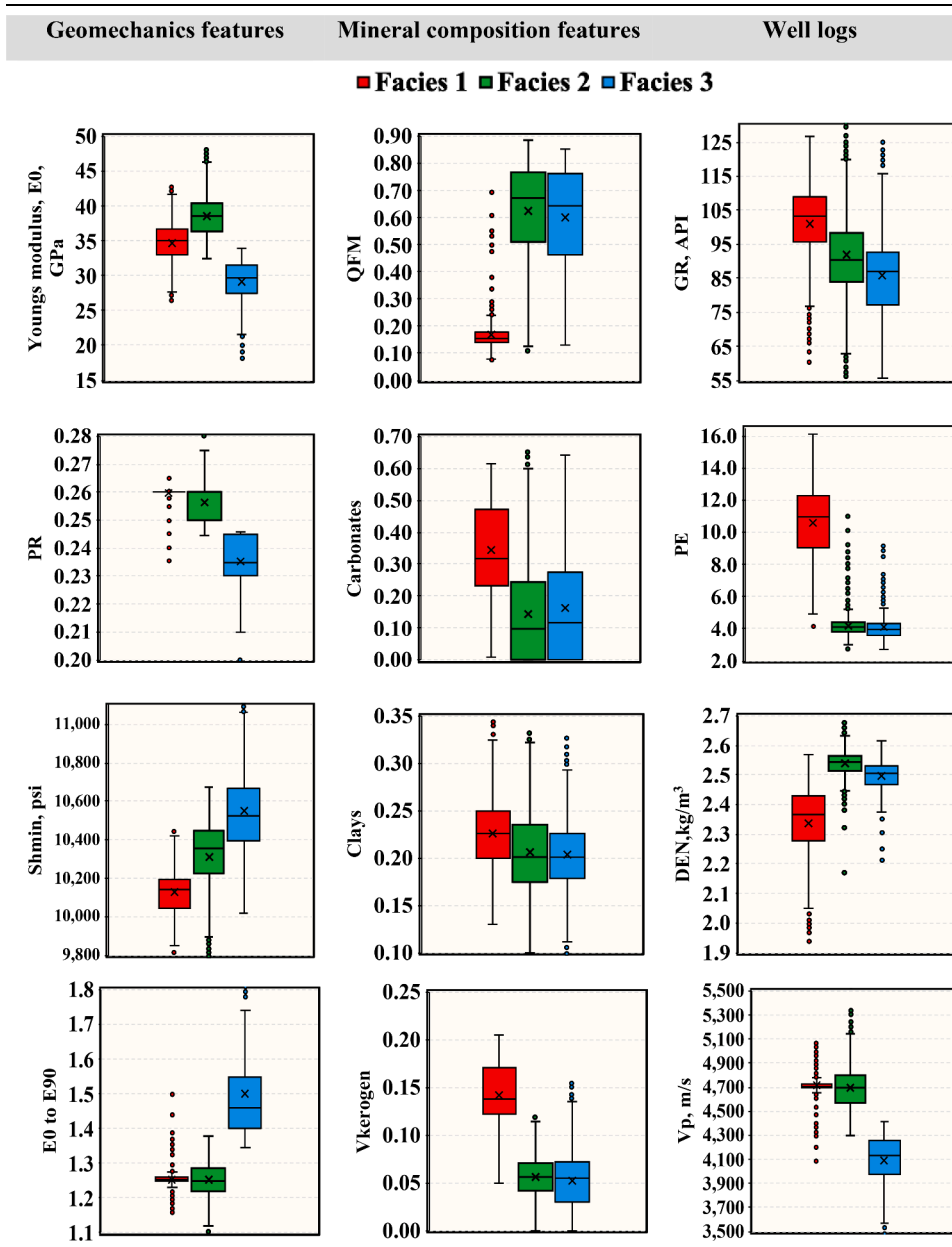
The confusion matrix plot is a commonly used technique describing the classification performance. It is a tabular summary of the number of correct and incorrect predictions made by a classifier. The better performance has more numbers on the diagonal grids of the confusion matrix. Even though we do not know the true values of the clusters, we incorporate confusion matrix as a performance evaluator in our

workflow to do a hybrid-cross-validate matrix of prediction performance from two clustering methods for both the training dataset and the test dataset. Table 3 and Fig. 11 give the cross-validate results of both the training dataset and the test dataset. Facies 1 prediction yields promising agreement from both clustering methods. As for facies 2, it includes most of the data population and preserves the most mismatching between two clustering methods. Facies 3 has about the same number of data from two methods. Among all three facies, we reached 90.75% and 90.58% confidence of the facies identifying from the cross-validation process of the training set and test set, respectively.

4 shows the grid matrices of boxplots of geomechanics features, mineral composition features, and well log features from k -means clustering. The boxplots of three facies from each feature indicate the distributions of that facies. Recalling the objective of this step, we are investigating the fracability of rock once the rock facies or the geomechanics facies is obtained. Through the cross-validation, the results show that clustering performance of the distinguishes the facies apart as expected. Through the cross-validation, facies 1 has the intermediate horizontal Young’s modulus (E_0) range, highest Poisson’s ratio, lowest minimum horizontal stress range (Sh_{min}), and the intermediate elastic anisotropy (E_{0toE90}). Facies 2 has the highest horizontal Young’s modulus (E_0) range, intermediate Poisson’s ratio, intermediate minimum horizontal stress range (Sh_{min}) and the lowest elastic anisotropy (E_{0toE90}). Last, facies 3 has the lowest horizontal Young’s modulus (E_0) range, lowest Poisson’s ratio, highest horizontal stress range (Sh_{min}) and the highest elastic anisotropy (E_{0toE90}).

To further investigate the fracability, we built a geomechanics model in the fracture modeling software GOHFER to test the fracture propagation in selected 18 perforation depths in our investigated interval. Figure demonstrates the fracture geometries at each perforation marked as a cross along a vertical well. Figure a shows the minimum horizontal stress profile while Figure b is the predicted geomechanics facies from the clustering workflow. We have also summarized the simulated fracture gross half-length and effective half-length, fracture height, and fracture flowing area of each landing points in Fig. 13 a, Fig. 13 b, and Fig. 13 c in bar plots, respectively. Fig. 12 and Fig. 13 indicate that L2, L3, L6, and L7 grows relatively short fractures while their geometries are smoother. Also, these integrals correspond to the facies 1 (marked in red) In these regions, the vertical profile is relatively constant in terms of both minimum horizontal stress and the rock facies. In other words, the

Table 4

The box plot grid matrix of geomechanics facies from *k-means*.

geomechanical facies and other geomechanical properties show relatively low heterogeneity. However, the E0toE90 attributes indicate this group (facies 1 in red) has intermediate anisotropy behavior while it also has the highest TOC (labeled as “Vkerogen” in Table 4). This means facies 1 is enriched with organic content. However, when checking the water saturation profile, facies 1 has nearly 100% of water saturation, which means the oil saturation is minimal. This would possibly be because of the lack of favorable maturity and thermal condition. In other words, the solid organic content (kerogen) has not gone through the thermal maturation to become liquid hydrocarbons.

In contrast, for facies 3 (marked in blue) corresponds to the high minimum horizontal stress, low Young's modulus facies, highest E0toE90 ratio, and intermediate-low kerogen volume fraction. These facies would not be the most favorable in the fracturing perspective, especially when the minimum horizontal stress is very high. It takes relatively higher energy to cause the tensile failure of the rock through hydraulic fracturing. In fact, from Figure b, fracture geometries

generated in L11-13 indicate the height growth restrictions on vertical propagation direction by the presence of facies 3. Perforations at L8 to L10 show reduced height growth. The facies in these intervals have shown the frequent alterations among facies 1, facies 2 and facies 3. Usually, when rock properties undergo drastic changes as above-mentioned situation, the hydraulics would show several small-scaled “breakdown” from the wellhead pressure curve instead of an obvious peak showing the breakdown of the rock. Practically, this will lead to longer duration and more difficulties as compared to an ideal fracturing job. This frequent change of rock properties is usually referred to the high heterogeneity of the rock, which is highly not favorable.

Layer L14-L18 mainly contains the facies 2, mixed with a small amount of facies 3—fracture generated in layer L14 to L18 show relatively promising fracture height and length. Although the fracture height grows are constrained by some strikes of mixed facies 3 in the facies 2 intervals, the continuity of facies 2 ensures the fracture propagation as the fracture geometries yield better results. Among L14-L18,

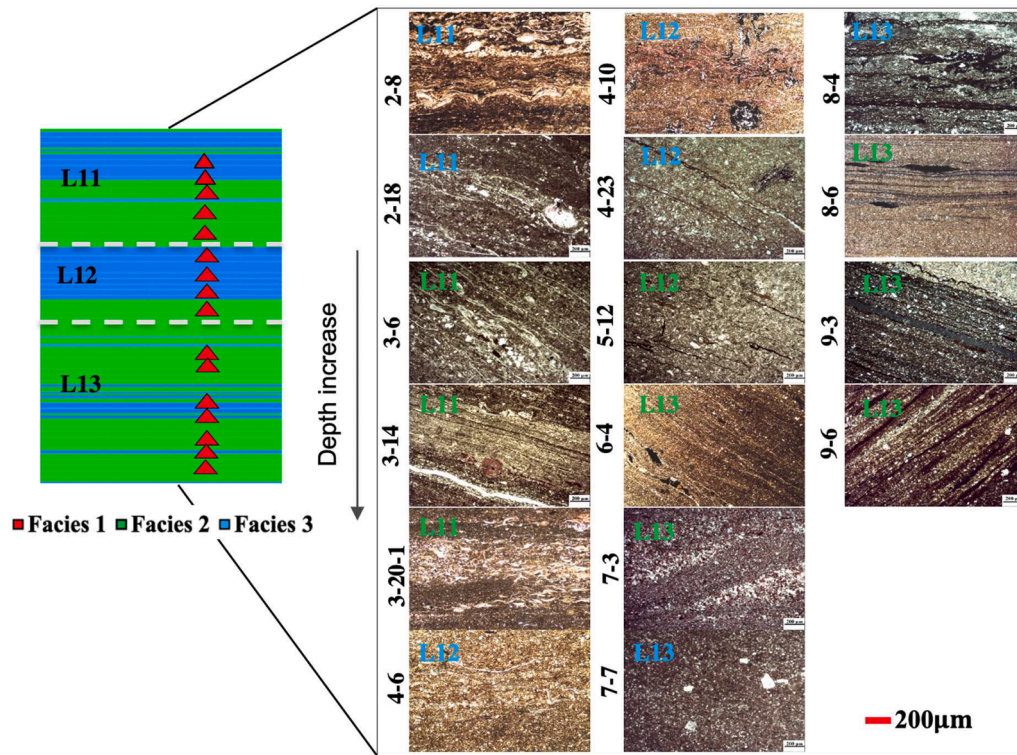


Fig. 14. The microstructure information relevant to the geomechanical facies. Each red triangle mark in the facies profile indicates a depth where the microstructure image was taken from. The microstructure image is marked to specific layer. The color of the layer in the image shows the facies color it belongs to. Herein, the green refers to facies 2 while the blue refers to facies 3.

L14 and L15 fall in the 100% water saturation interval. From the geomechanics perspective, L14 and L15 are favored while it is not considered a pay zone.

Typically, the presence of clay minerals impairs the mechanical behavior of the rock and introduces anisotropy due to their crystallographic character. The degree of alignment of clay is dominated by the depositional environment, clay composition, and the formation's thermal maturity [7],[21]. The bedding plane orientations are usually formed when the clay content gathers together with other rock matrix components in a preferred direction (typically horizontal) in the micro to centi-meter range. The preferred orientation from clay aggregation can disappear when its composition is less than 30% of the total volume [3],[27].

The microfracture hardly preserves the voids but is usually filled by the organic content or fines/cementing materials [12]. According to Zoback and Kohli [28], in organic-rich rock, the pore types largely involve organic matter pores and clay involved inter-particle porosity, preferentially laying in the bedding plane direction. In other words, there will be a dominant or a narrowed dominant pore orientation range in organic-rich reservoir rocks. In contrast, the siliciclastic rocks with less clay mineral composition would have a large portion of intraparticle particle or intra-granular pore types with no preferred pore orientations. Thus, the mineral compositions, microstructure related pore types highly inter-relate to the rock mechanical and its anisotropy mechanical behaviors.

To evaluate the effects of microstructures, including clays and beddings in the microscopic scope to the fracability of the rock, we have selected microstructure thin-section images of which depths are marked in red triangles from the landing zone 11 to zone 13 at aforementioned vertical section (Figure). Microstructure images descriptions interpreted from lab results show laminar or clear stratified sedimentary rocks (either felsic fine-grained, dolomitic fine-grained, limestone fine-grained, or mixed fine-grained). For the facies 3, it is mainly described as bulky or massive fine-grained felsic sedimentary rock (limestonic,

analcite-bearing felsic, or mixed). In Figure , the main difference is that facies 3 (blue) has less significant stratified character as facies 3 show slightly or evident laminations. On the macroscopic scale, it essentially leads to the rock mechanical behaviors. Some facies 3 layers preserve a great amount of solid kerogen bearing content from the images while the mineral composition does not show obvious lamination. The labeled images 2-18, 3-6, 5-12, 7-3, 9-3 are recorded preserving the microfractures. These fractures are on the scale of several hundred microns to several millimeter. Four out of five of them are from facies 2 which shows less mechanical strength. This indicates that the microfractures are more commonly-developed in stratified or laminated rocks. It also shows the lowest rock strength may be observed when the rock develops microfractures. (Fig. 14)

4. Conclusions

We have investigated the capabilities of various machine learning algorithms to unveil the connection between rock attributes and elastic properties. The following summaries highlight the principal findings:

- Non-linear regression methods, such as MARS and Artificial Neural Networks (ANN), demonstrate more promising predictive capabilities compared to linear regularized methods (including LASSO, Ridge, and Elastic Net) and are recommended in cases where linearity proves elusive during multivariate regression analyses.
- Microstructural insights garnered from thin-section images of rocks strongly correspond with classified geomechanical / fracability facies. An alignment is observed between the brittleness and anisotropy characteristics and the inherent nature of shale rock. This observation can be harnessed to determine optimal landing zones in vertical fracturing sensitivity studies based on geomechanical facies outcomes.
- The degree of fracturability is tightly linked to the continuous presence of geomechanical facies within a given interval during fracture

propagation, whereas the specific geomechanical facies are more closely associated with fracture initiation.

In essence, our exploration of machine learning algorithms sheds light on the intricate relationship between rock properties and elastic behavior. The insights garnered from these findings hold potential implications for optimizing hydraulic fracturing strategies and understanding shale rock behavior.

CRedit authorship contribution statement

Yiwen Gong: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Ilham El-Monier:** Writing – review & editing, Supervision, Project administration. **Mohamed Mehana:** Writing – review & editing, Supervision, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that there is no interest conflict.

Data availability

The data that has been used is confidential.

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